

DECISION TREES

Decision Trees are considered to be one of the most popular approaches for representing classifiers. Researchers from various disciplines such as statistics, machine learning, pattern recognition, and Data Mining have dealt with the execution of growing a decision tree from available data.

Tree models where the target variable can take a discrete set of values are called **classification trees**;

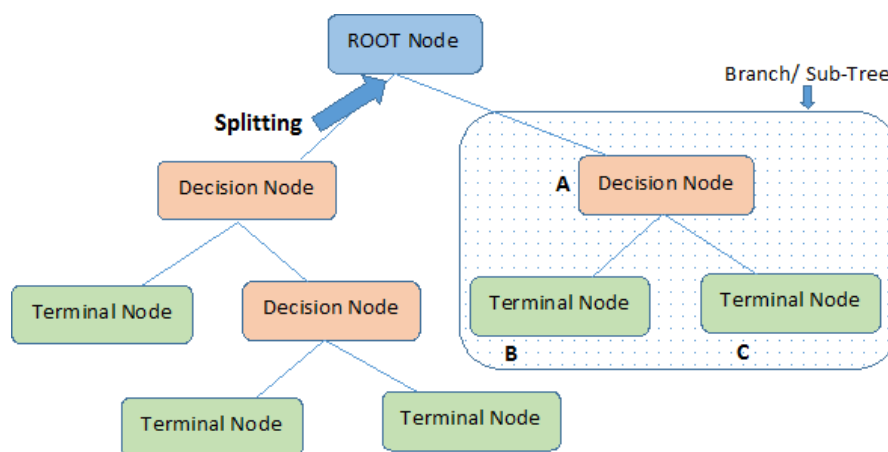
in these tree structures, leaves represent class labels and branches represent conjunctions of features that lead to those class labels.

Decision trees where the target variable can take continuous values (typically real numbers) are called **regression trees**.

The term **Classification and Regression Tree (CART) analysis** is an umbrella term used to refer to both of the above procedures

Important Terminology related to Decision Trees

- **Root Node:** It represents entire population or sample and this further gets divided into two or more homogeneous sets.
- **Splitting:** It is a process of dividing a node into two or more sub-nodes.
- **Decision Node:** When a sub-node splits into further sub-nodes, then it is called decision node.
- **Leaf/ Terminal Node:** Nodes do not split is called Leaf or Terminal node.



Note:- A is parent node of B and C.

Pruning: When we remove sub-nodes of a decision node, this process is called pruning. You can say opposite process of splitting.

Branch / Sub-Tree: A sub section of entire tree is called branch or sub-tree.

Parent and Child Node: A node, which is divided into sub-nodes is called parent node of sub-nodes whereas sub-nodes are the child of parent node.

The primary challenge in the decision tree implementation is to identify which attributes do we need to consider as the root node and each level. Handling this is know the attributes selection. We have different attributes selection measure to identify the attribute which can be considered as the root note at each level.

The popular attribute selection measures:

- Information gain
- Gini index

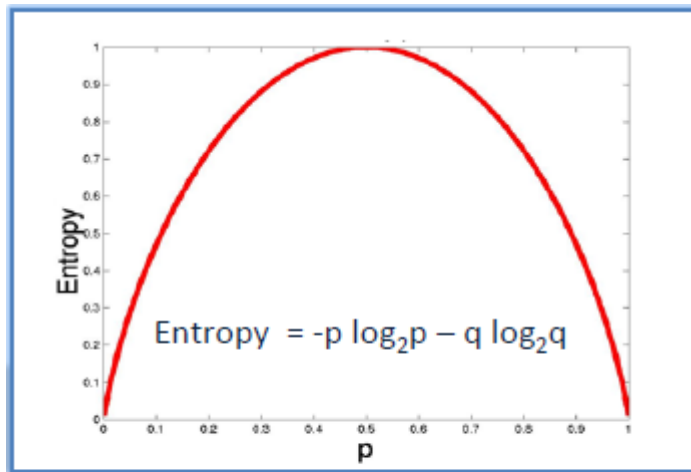
If dataset consists of “n” attributes then deciding which attribute to place at the root or at different levels of the tree as internal nodes is a complicated step. By just randomly selecting any node to be the root can’t solve the issue. If we follow a random approach, it may give us bad results with low accuracy.

For solving this attribute selection problem, researchers worked and devised some solutions. They suggested using some *criterion* like **information gain, gini index**, etc. These criterions will calculate values for every attribute. The values are sorted, and attributes are placed in the tree by following the order i.e, the attribute with a high value(in case of information gain) is placed at the root.

While using information Gain as a criterion, we assume attributes to be **categorical**, and for gini index, attributes are assumed to be **continuous**.

Entropy

A decision tree is built top-down from a root node and involves partitioning the data into subsets that contain instances with similar values (homogenous). ID3 algorithm uses entropy to calculate the homogeneity of a sample. If the sample is completely homogeneous the entropy is zero and if the sample is an equally divided it has entropy of one.



$$Entropy = -0.5 \log_2 0.5 - 0.5 \log_2 0.5 = 1$$

To build a decision tree, we need to calculate two types of entropy using frequency tables as follows:

- a) Entropy using the frequency table of one attribute:

$$E(S) = \sum_{i=1}^c -p_i \log_2 p_i$$

Play Golf	
Yes	No
9	5

$$\begin{aligned} Entropy(PlayGolf) &= Entropy(5,9) \\ &= Entropy(0.36, 0.64) \\ &= -(0.36 \log_2 0.36) - (0.64 \log_2 0.64) \\ &= 0.94 \end{aligned}$$

- b)

$$E(T, X) = \sum_{c \in X} P(c)E(c)$$

		Play Golf		
		Yes	No	
Outlook	Sunny	3	2	5
	Overcast	4	0	4
	Rainy	2	3	5
				14



$$\begin{aligned}
 E(\text{PlayGolf}, \text{Outlook}) &= P(\text{Sunny}) * E(3,2) + P(\text{Overcast}) * E(4,0) + P(\text{Rainy}) * E(2,3) \\
 &= (5/14) * 0.971 + (4/14) * 0.0 + (5/14) * 0.971 \\
 &= 0.693
 \end{aligned}$$

INFORMATION GAIN

Information Gain

The information gain is based on the decrease in entropy after a dataset is split on an attribute. Constructing a decision tree is all about finding attribute that returns the highest information gain (i.e., the most homogeneous branches).

Step 1: Calculate entropy of the target.

$$\begin{aligned}
 \text{Entropy}(\text{PlayGolf}) &= \text{Entropy}(5,9) \\
 &= \text{Entropy}(0.36, 0.64) \\
 &= -(0.36 \log_2 0.36) - (0.64 \log_2 0.64) \\
 &= 0.94
 \end{aligned}$$

Step 2: The dataset is then split on the different attributes. The entropy for each branch is calculated. Then it is added proportionally, to get total entropy for the split. The resulting entropy is subtracted from the entropy before the split. The result is the Information Gain, or decrease in entropy.

		Play Golf	
		Yes	No
Outlook	Sunny	3	2
	Overcast	4	0
	Rainy	2	3
Gain = 0.247			

		Play Golf	
		Yes	No
Temp.	Hot	2	2
	Mild	4	2
	Cool	3	1
Gain = 0.029			

		Play Golf	
		Yes	No
Humidity	High	3	4
	Normal	6	1
Gain = 0.152			

		Play Golf	
		Yes	No
Windy	False	6	2
	True	3	3
Gain = 0.048			

Step 3: Choose attribute with the largest information gain as the decision node, divide the dataset by its branches and repeat the same process on every branch.

		Play Golf	
		Yes	No
Outlook	Sunny	3	2
	Overcast	4	0
	Rainy	2	3
Gain = 0.247			

		Outlook	Temp	Humidity	Windy	Play Golf
Outlook	Sunny	Sunny	Mild	High	FALSE	Yes
		Sunny	Cool	Normal	FALSE	Yes
		Sunny	Cool	Normal	TRUE	No
		Sunny	Mild	Normal	FALSE	Yes
		Sunny	Mild	High	TRUE	No
	Overcast	Overcast	Hot	High	FALSE	Yes
		Overcast	Cool	Normal	TRUE	Yes
		Overcast	Mild	High	TRUE	Yes
		Overcast	Hot	Normal	FALSE	Yes
	Rainy	Rainy	Hot	High	FALSE	No
		Rainy	Hot	High	TRUE	No
		Rainy	Mild	High	FALSE	No
		Rainy	Cool	Normal	FALSE	Yes
		Rainy	Mild	Normal	TRUE	Yes

Temp	Humidity	Windy	Play Golf
Hot	High	FALSE	Yes
Cool	Normal	TRUE	Yes
Mild	High	TRUE	Yes
Hot	Normal	FALSE	Yes

